IOQM Grade 9 Number Theory

Divisibility, GCD, LCM & Euclidean Structure
Student Learning Pack

HOW TO USE THIS PACK

Work in this order: RECONNECT -> DISCOVER -> MAKE SENSE -> TRY -> DIAGNOSE -> FADE -> ADOPT -> TRANSFER.
At TRY, attempt before reading help. During fading, support decreases H3 -> H2 -> H1 -> H0. The final H0 section gives no method labels.

1. RECONNECT - WHAT YOU ALREADY KNOW

You may already compute HCF/LCM and use divisibility tests. The new goal is method selection. Before calculating, ask: Am I looking for a divisor or a multiple? Can subtraction expose a common divisor? Can Euclid shrink the numbers? Is the condition really about a prescribed remainder?

Quick diagnostic - write only the first idea.
1) Greatest integer dividing 173 and 239 with the same remainder.
2) Least positive integer divisible by 12, 15 and 18.
3) gcd(987,610) efficiently.
4) gcd(a,b)=6 and lcm(a,b)=180: what invariant is immediate?

Expected first ideas: difference; lcm; Euclid; gcd*lcm=product.

2. DISCOVER - DIVISIBILITY IS AN EQUATION

For nonzero integer a, a|b means b=ak for some integer k.
This is stronger than a symbol to recognize. It lets us combine divisible quantities.
If d|A and d|B, then d|(mA+nB) for all integers m,n.
So common divisors survive subtraction and integer linear combinations.

CONTRAST A - TEST OR STRUCTURE?

Question: Is the task merely to decide whether one displayed number is divisible, or must one divisor remain valid after numbers are combined?
- A visible digit pattern may invite a divisibility test.
- A common-divisor relation should usually be translated into equations and combined.

Example: if d|(7n+5) and d|(4n+3), then d divides 4(7n+5)-7(4n+3)=-1. Hence d=1.

3. MAKE SENSE - EUCLID PRESERVES GCD

If d divides both a and b, then d divides a-qb for every integer q. Conversely, a common divisor of b and a-qb also divides a.
Therefore gcd(a,b)=gcd(b,a-qb).
Choose q so the new number is small. The Euclidean algorithm repeats this until the remainder is 0.

Example: gcd(987,610)
987=1*610+377
610=1*377+233
377=1*233+144
233=1*144+89
144=1*89+55
89=1*55+34
55=1*34+21
34=1*21+13
21=1*13+8
13=1*8+5
8=1*5+3
5=1*3+2
3=1*2+1

2=2*1+0
So gcd=1.

CONTRAST B - EUCLID OR FACTOR EVERYTHING?

Both can be correct. Euclid is usually cheaper when subtraction/remainders shrink the pair quickly. Factorization belongs only when factor structure is itself useful.

4. MAKE SENSE - SAME REMAINDER MEANS DIFFERENCES

Suppose a and b leave the same remainder r on division by positive d:
a=dq+r, b=dp+r, with 0<=r<d.
Subtract: a-b=d(q-p). Therefore d|(a-b).
For several numbers, every valid divisor must divide all relevant differences. The greatest valid divisor is the gcd of those differences, after checking the remainder condition.

Example: greatest integer dividing 211, 323 and 435 with the same remainder.
Differences: 112 and 112. Greatest candidate d=112.
Check: 211=1*112+99, 323=2*112+99, 435=3*112+99. Answer 112.

CONTRAST C - SAME REMAINDER, TWO DIFFERENT TARGETS

- Greatest unknown divisor leaving equal remainders -> subtract -> gcd of differences.
- Least number with a prescribed remainder modulo several divisors -> remove the remainder -> common multiple -> lcm.

The words "same remainder" do not choose the method by themselves. The target does.

5. MAKE SENSE - LCM IS THE LEAST SYNCHRONIZATION POINT

A common multiple of a,b is divisible by both. lcm(a,b) is the least positive one.
Typical cues: least positive integer divisible by..., first time together..., smallest number satisfying all divisibility conditions.
Example: cycles of 18, 24 and 30 minutes meet again after lcm(18,24,30)=360 minutes.

Prescribed remainder example: least N>7 leaving remainder 7 on division by 12,18,30.
Then N-7 is a common multiple. Least choice: N-7=lcm(12,18,30)=180, so N=187.

CONTRAST D - GCD OR LCM?

Ask what the answer itself must be.
- A divisor of several quantities -> gcd direction.
- A number divisible by several quantities -> lcm direction.

Do not choose from the size of the numbers or from a memorized keyword alone.

6. MAKE SENSE - GCD AND LCM RECONSTRUCTION

For positive integers a,b:
gcd(a,b)*lcm(a,b)=ab.
If g=gcd(a,b), write a=gu, b=gv with gcd(u,v)=1.
Then lcm(a,b)=guv. This normalization separates the common part g from the coprime parts u,v.

Example: gcd(a,b)=12, lcm(a,b)=420.
ab=12*420=5040.
If the pair itself is requested, write a=12u,b=12v. Since lcm=12uv=420, uv=35 and gcd(u,v)=1.
The unordered coprime pairs are (1,35),(5,7).

7. MAKE SENSE - DIVISIBILITY CHAINS

If a|b and b|c, then a|c. If a common divisor divides several expressions, combine them to force a smaller fixed number.
Example: d|(4n+7) and d|(7n+13). Then d divides 7(4n+7)-4(7n+13)=-3. So d is a positive divisor of 3. Check attainability before declaring all candidates.

8. TRY - ATTEMPT BEFORE HELP

Problem A: Find the greatest integer dividing 221,323,425 with the same remainder.
Try before help.

H3 execution: compute differences and take their gcd.
H2 structure: a common remainder disappears after subtraction.
H1 recognition: is the unknown a divisor or a multiple?
H0: Find the greatest integer dividing 437,581,725 with the same remainder.

Problem B: Find the least positive integer divisible by 18,24,30.
Try before help.
H3 execution: build the lcm.
H2 structure: the answer must be a common multiple.
H1 recognition: is the target least or greatest, divisor or multiple?
H0: Find the least N>9 leaving remainder 9 on division by 12,15,20.

9. DIAGNOSE - TEMPTING WRONG STARTS

Error 1: "Same remainder means lcm." Repair: first identify whether the unknown is a divisor or the number being reconstructed.
Error 2: "Largest number means gcd." Repair: largest/greatest matters only after identifying what quantity is being optimized.
Error 3: Factor everything immediately. Repair: test whether Euclid or subtraction reduces the structure faster.
Error 4: Use gcd*lcm=ab without positivity or without checking pair constraints. Repair: state the domain and normalize if a,b are required.
Error 5: List divisors but do not check attainability. Repair: substitute back into the original condition.
Error 6: Confuse digit divisibility tests with structural divisibility. Repair: tests answer yes/no; structural reasoning preserves a divisor through algebraic combinations.

10. ADOPT - FIRST-MOVE ROUTER

TARGET -> DIVISOR OR MULTIPLE? -> DIFFERENCE/REDUCTION? -> GCD OR LCM? -> CHECK.

Use these automatic first lines:
- a|b -> b=ak, k integer.
- same unknown divisor + equal remainders -> d|(a-b).
- gcd of large pair -> a=qb+r, then gcd(a,b)=gcd(b,r).
- least simultaneous divisibility -> N=lcm(...), or N-r=lcm(...) for prescribed remainder r.
- gcd and lcm both given -> gcd*lcm=ab; if pair needed, a=gu,b=gv,gcd(u,v)=1.
- common divisor of linear expressions -> form an integer linear combination that eliminates a variable.

11. FIRST-STEP REFERENCE - CLOSE CONTRASTS

A) "Greatest divisor leaving same remainder" vs "least number leaving remainder r": first uses differences/gcd; second removes r and uses lcm.
B) "Does 7425 divide by 9?" vs "d divides two expressions": first may use a digit test; second uses structural divisibility.
C) "Compute gcd of two large numbers" vs "count divisors / compare prime exponents": Euclid belongs here; prime-exponent canon belongs to NT-03.
D) "Residue cycle of a huge power" belongs to NT-02. NT-01 supplies only the divisibility meaning needed downstream.

30-second check before computing:
1) What is requested?
2) Is the answer itself a divisor or a multiple?
3) Can subtraction remove a repeated remainder?
4) Can Euclid shrink the pair?
5) Is a prescribed remainder attached to the number being reconstructed?
6) What final condition must be checked?

12. RECOGNITION AND FIRST-LINE LAB

Write only the first useful mathematical line.

1) Greatest integer dividing 188 and 296 with the same remainder.
2) Least positive integer divisible by 14,20,35.
3) Compute gcd(12345,5432) efficiently.
4) gcd(a,b)=9, lcm(a,b)=252. Write an invariant involving ab.
5) d divides 5x+7y and 2x+3y. Write a useful integer combination.
6) N gives remainder 4 on division by 6,10,15. First line for least N>4.
7) Greatest divisor leaving the same remainder for 812,1046,1280. Write the reduction only.
8) Alarms ring every 18 and 24 minutes. Write the operation for the next common time.
9) a=gu,b=gv with g=gcd(a,b). What condition must u,v satisfy?
10) Decide whether 84|1260 by translating to an integer quotient.

13. PRACTICE AND TRANSFER

1) Verify 18|234 by quotient form.
2) Find gcd(84,126) and lcm(84,126).
3) Find the greatest integer dividing 155 and 239 with the same remainder.
4) Find the least positive integer divisible by 16,20,24.
5) Use Euclid to compute gcd(2025,748).
6) Find the greatest divisor leaving the same remainder for 437,581,725.
7) Find the least N>9 leaving remainder 9 on division by 12,15,20.
8) gcd(a,b)=12 and lcm(a,b)=420. Determine ab.
9) d divides 7x+5y and 3x+2y. Form combinations and constrain d when gcd(x,y)=1.
10) Three machines reset every 18,24,40 minutes from 09:00. When next together?
11) Find the greatest d such that 1001,1457,1913 leave the same remainder on division by d.
12) Suppose d|(4n+7) and d|(7n+13). Determine all possible positive d, including attainability.

TRANSFER

T2 representation: Explain why Euclid and equal-remainder subtraction are the same gcd-invariance idea.
T3 context: A circular track has markers every 84 m and 126 m from a common origin. What spacing question creates a gcd rather than an lcm?
T3 changed target: A machine cycle problem asks for the greatest common calibration interval instead of the next simultaneous event. Explain why the first move changes.
T4 downstream bridge: Rewrite "a and b leave the same remainder on division by d" as d|(a-b). This is the prerequisite form NT-02 may retrieve before introducing congruence notation.
WHY-NOT: For gcd(2025,748), explain why full prime factorization may be valid but oversized. For a prescribed-remainder reconstruction, explain why taking gcd of the divisors answers the wrong target.

14. VALIDATED IOQM ANCHORS

IOQM-2025-Q02: divisibility-counting anchor. Independent computation gives floor(100/3)-floor(100/6)=33-16=17, agreeing with the final official key.
IOQM-2025-Q27: lcm/gcd-normalization anchor. Independent reduction and admissible-case count gives 40, agreeing with the final official key.
Exact historical wording stays with the validated paper; this pack keeps stable year/question provenance without pretending author-created items are PYQs.

15. H0 MASTERY TEST - NO METHOD LABELS

Part A - Notice and first useful line

1) Greatest positive integer leaving the same remainder when dividing 428,596,764.
2) Least positive integer >11 leaving remainder 11 on division by 18,24,30.
3) gcd(123456,7890) without complete prime factorization.

4) gcd(a,b)=14, lcm(a,b)=840. Write the first invariant if ab is requested.

Part B - Full mixed solve

5) d divides 11n+8 and 7n+5. Find the largest possible d for some integer n.
6) Find all positive d for which 305,473,641 leave the same remainder on division by d.
7) lcm(a,b)=360, gcd(a,b)=12, one number is 72. Find the other.
8) Events recur every 28,42,63 days and occur together today. When next together?
9) Prove gcd(a,b)=gcd(a,a-b).
10) Explain why equal remainders on division by d force d|(a-b).

Part C - Same surface, different decision

11) Problem A asks for the greatest divisor leaving remainder 5 on several numbers. Problem B asks for the least number leaving remainder 5 on several divisors. Write the different first line for each.
12) One question asks whether 7425 is divisible by 9; another says d divides two linear expressions. Why can the first use a digit test while the second needs structural divisibility?

Part D - Transfer and WHY-NOT

13) Markers are placed at distances 84 m,126 m,210 m from an origin. Formulate and solve a greatest-spacing question using this topic.
14) A number N leaves remainder 7 when divided by 12,18,30. Find the least N>7 and state why gcd is the wrong operation.
15) Explain why factoring both numbers completely is not the cheapest first move for gcd(123456,7890), even though it can work.
16) A learner writes "same remainder -> lcm" without reading the target. Give a counterexample pair of problems showing why that rule fails.

MASTERY REFLECTION

For any three items, record: What clue triggered the first move? What nearby method was tempting? Why was it less direct or wrong? What final check confirms the result?